SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	**Total Marks:**	100
CO2 Statement:	*Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)*		
Student Name:	Ganji Madhan kumar	**Reg. No.:**	192472328
Section / Batch:		**Date:**	18-07-2026

Instructions to Students

- Answer ALL questions. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type		Questions	Marks
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1 Q2,Q3	40 60 (2X30)
	GRAND TOTAL:	**100 Marks**	

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I ______________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ______________________

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (40 Marks)						
Q2 (30 Marks)						
Q3 (30 Marks)						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ______________	Signature: ______________	Signature: ______________
Date: ______________	Date: ______________	Date: ______________

CO2-AT01

- Analytical Problem Solving

Name: G.Madhan kumar
Reg No: 192472328
Code : CSA0624
Course: DAA

(1) Selection Sort

[64, 25, 12, 22, 11].

Given array

64	25	12	22	11

Pass 1: Smallest element → 11

Swap 64 and 11

11	25	12	22	64

Comparisons = 4

Pass 2: remaining array

25	12	22	64

Small element = 12

swap 25 and 12

11	12	25	22	64

Comparisons = 3

Pass 3: remaining array

25	22	64

Smallest element = 22

Swap 25 and 22

11	12	22	25	64

Comparisons = 2

Pass 4: remaining array

25	64

Smallest element = 25

No need to swap

11	12	22	25	64

Comparisons = 1

Total Comparisons

$4+3+2+1 = 10$

Final Sorted Array

11	12	22	25	64

Time Complexity

Case	Complexity
Best	$O(n^2)$
Average	$O(n^2)$
Worst	$O(n^2)$

② Closest Pair - Brute Force Analysis

Given points

(2,3), (12,30), (40,50), (5,1), (12,10), (3,4)

Distance formula $d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$

Distance calculation:

$P_1 = (2,3)$ & $P_2 = (12,30)$

$= \sqrt{(12-2)^2 + (30-3)^2}$

$= \sqrt{10^2 + 27^2}$

$= \sqrt{100+729}$

$= \sqrt{829}$

$= 28.79$

P_1 & P_3 $\quad P_1 = (2,3)$ & $P_3 = (40,50)$

$= \sqrt{(40-2)^2 + (50-3)^2}$

$= \sqrt{38^2 + 47^2}$

$= \sqrt{1444 + 2209}$

$= \sqrt{3653}$

$= 60.44$

P_1 & P_4 $\quad P_1 = (2,3)$ & $P_4 = (5,1)$

$= \sqrt{(5-2)^2 + (1-3)^2}$

$= \sqrt{3^2 + (-2)^2}$

$= \sqrt{9+4}$

$= \sqrt{13}$

$= 3.60$

P_1 & P_5 $\quad P_1 = (2,3)$ & $P_5 = (12,10)$

$= \sqrt{(12-2)^2 + (10-3)^2}$

$= \sqrt{10^2 + 7^2}$

$= \sqrt{100+49}$

$= \sqrt{149}$

$= 12.20$

P_1 & P_6 $\quad P_1 = (2,3)$ & $P_6 = (3,4)$

$= \sqrt{(3-2)^2 + (4-3)^2}$

$= \sqrt{1^2 + 1^2}$

$= \sqrt{2}$

$= 1.41$

P_2 & P_3 - $P_2 = (12,30)$ & $P_3 = (40,50)$

$= \sqrt{(40-12)^2 + (50-30)^2}$

$= \sqrt{28^2 + 20^2}$

$= \sqrt{784 + 400}$

$= \sqrt{1184}$

$= 34.40$

P_2 & P_4 $\rightarrow P_2 = (12,30)$ & $P_4 = (5,1)$

$= \sqrt{(5-12)^2 + (1-30)^2}$

$= \sqrt{(-7)^2 + (-29)^2}$

$= \sqrt{49 + 841}$

$= \sqrt{890}$

$= 29.15$

P_2 & P_5 $\rightarrow P_2 = (12,30)$ & $P_5 = (12,10)$

$= \sqrt{(12-12)^2 + (10-30)^2}$

$= \sqrt{0^2 + (-20)^2}$

$= \sqrt{0+400}$

$= \sqrt{400}$

$= 20$

P_2 & P_6 $\rightarrow P_2 = (12,30)$ & $P_6 = (3,4)$

$= \sqrt{(3-12)^2 + (4-30)^2}$

$= \sqrt{(-9)^2 + (-26)^2}$

$= \sqrt{81 + 676}$

$= \sqrt{757}$

$= 27.31$

P_3 & P_4 → $P_3 = (40, 50)$, $P_4 = (5, 1)$

$= \sqrt{(5-40)^2 + (1-50)^2}$

$= \sqrt{(-35)^2 + (-49)^2}$

$= \sqrt{1225 + 2401}$

$= \sqrt{3626}$

$= 59.03$

P_3 & P_5 → $P_3 = (40, 50)$, $P_5 = (12, 10)$

$= \sqrt{(12-40)^2 + (10-50)^2}$

$= \sqrt{(-28)^2 + (-40)^2}$

$= \sqrt{784 + 1600}$

$= \sqrt{2384}$

$= 48.83$

P_3 & P_6 → $P_3 = (40, 50)$, $P_6 = (3, 4)$

$= \sqrt{(3-40)^2 + (4-50)^2}$

$= \sqrt{(-37)^2 + (-46)^2}$

$= \sqrt{1369 + 2116}$

$= \sqrt{3485}$

$= 58.69$

P_4 & P_5 → $P_4 = (5, 1)$, $P_5 = (12, 10)$

$= \sqrt{(12-5)^2 + (10-1)^2}$

$= \sqrt{7^2 + 9^2}$

$= \sqrt{49 + 81}$

$= \sqrt{130}$

$= 11.40$

P_4 & P_6 → $P_4 = (5, 1)$, P_6

$= \sqrt{(3-5)^2 + (4-1)^2}$

$= \sqrt{(-2)^2 + 3^2}$

$= \sqrt{4+9}$

$= \sqrt{13}$

$= 3.61$

P_5 & P_6 → $P_5 = (12, 10)$, $P_6 = (3, 4)$

$= \sqrt{(3-12)^2 + (4-10)^2}$

$= \sqrt{(-9)^2 + (-6)^2}$

$= \sqrt{81 + 36}$

$= \sqrt{117}$

$= 9.85$

Closest pair

(2, 3) and (3, 4)

Distance = 1.41

Number of Comparisons

– there are 6 points

$n(n-1)/2$

$= 6(6-1)/2$

$= 6(5)/2$

$= 30/2$

= 15 Comparisons

Time Complexity

Best = $O(n^2)$

Average = $O(n^2)$

Worst = $O(n^2)$

Convex Hull - Brute Force Approach

Given points

(0,3), (2,2), (1,1), (2,1), (3,0), (0,0), (3,3)

Concept of Convex Hull.

A convex hull is the smallest convex polygon that contains all the given points.

Hull Edges:

Checking every possible line pair, the valid boundary edges are

(0,0) → (3,0)

(3,0) → (3,3)

(3,3) → (0,3)

(0,3) → (0,0)

Interior points

(1,1)

(2,1)

(2,2) are inside the polygon.

Total Comparisons

Number of points $n=7$

possible pairs → $7 \times 6/2$

$= 21$ pairs.

Approximate comparisons → $21 \times 5 = 105$

Time complexity

$O(n^3)$

All points lie inside or on the polygon formed by

(0,0), (3,0), (3,3), (0,3)

Hence, the convex hull is correct.

④ Bubble Sort - Analytical Problem

Given Array

45	12	78	34	23

Pass 1:

12	45	78	34	23
12	45	78	34	23
12	45	32	78	23
12	45	32	23	78

Pass 2:

12	45	32	23	78
12	32	45	23	78
12	32	23	45	78

Pass 3:

12	32	23	45	78
12	23	32	45	75

Pass 4:

12	23	32	45	75

- Total Comparisons

$4+3+2+1=10$

- total swaps $3+2+1=6$

- Time Complexity.

Case	Complexity
Best	$O(n)$
Average	$O(n^2)$
Worst	$O(n^2)$

Linear Search - Analytical Problem

Data set →

23	45	12	67	34	89	10

Key → 34

Compare

$23 \neq 34$

Comparison 1

$45 \neq 34$ → Comparison 2

$12 \neq 34$ → Comparison 3

$67 \neq 34$ → Comparison 4

$34 = 34$ → Comparison 5

Element found at Index = 4, Position = 5

Total comparisons - 5

Time Complexity

Case	Complexity
Best	$O(1)$
Average	$O(n)$
Worst	$O(n)$

Pseudocode:

```
Linear search (A, key):
    for i = 0 to n-1:
        if A[i] == key:
            return i
    return -1
```